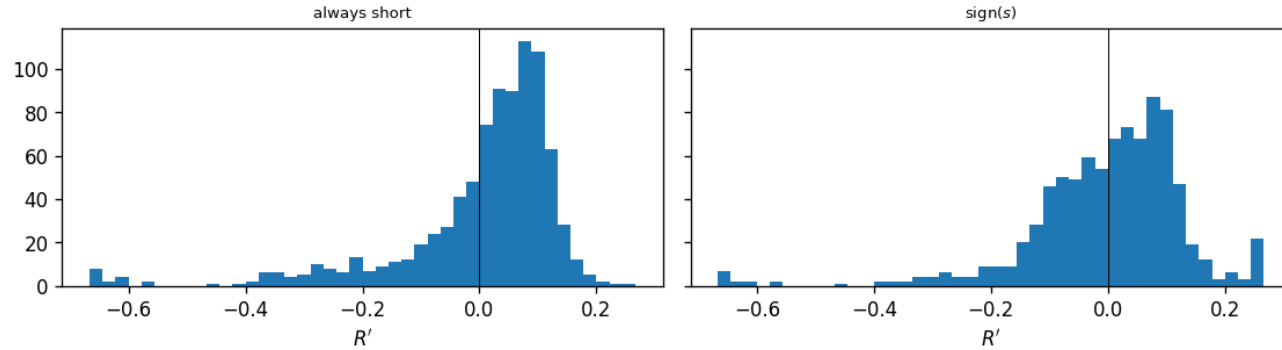


0DTE nearest-OTM package: rule table by strategy, every intraday entry window

Rule table by strategy — entry 10:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{clk}	n_{buy}	%buy	Sharpe _{clk} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	864	0.002	0.149	-1.12	0.25	-2.71	10.66	0.49	0.26	0	0.0	-3.74
<i>sign(s) (± 1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	864	0.002	0.149	-1.12	0.67	-1.73	10.56	0.36	0.19	307	35.5	-3.83
block-diagonal ridge	864	-0.001	0.149	-1.12	0.65	-1.76	10.42	-0.20	-0.11	303	35.1	-4.18
block-diagonal ridge, without the FOMC columns	864	0.001	0.149	-1.12	0.65	-1.60	10.51	0.19	0.11	312	36.1	-4.00
LightGBM [†]	864	-0.001	0.149	-1.12	0.67	-1.89	10.43	-0.17	-0.09	286	33.1	-4.16
XGBoost [†]	864	-0.003	0.149	-1.12	0.67	-1.88	10.33	-0.59	-0.32	281	32.5	-4.37
lasso (causally tuned) [†]	864	-0.003	0.149	-1.12	0.65	-1.72	10.34	-0.57	-0.31	310	35.9	-4.37
lasso ($\alpha = 10^{-4}$)	864	-0.005	0.149	-1.12	0.65	-1.88	10.25	-0.94	-0.51	294	34.0	-4.56
elastic net (causally tuned) [†]	864	-0.001	0.149	-1.12	0.65	-1.71	10.42	-0.21	-0.11	314	36.3	-4.17

block-diagonal ridge, midpoint fills: entry 10:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



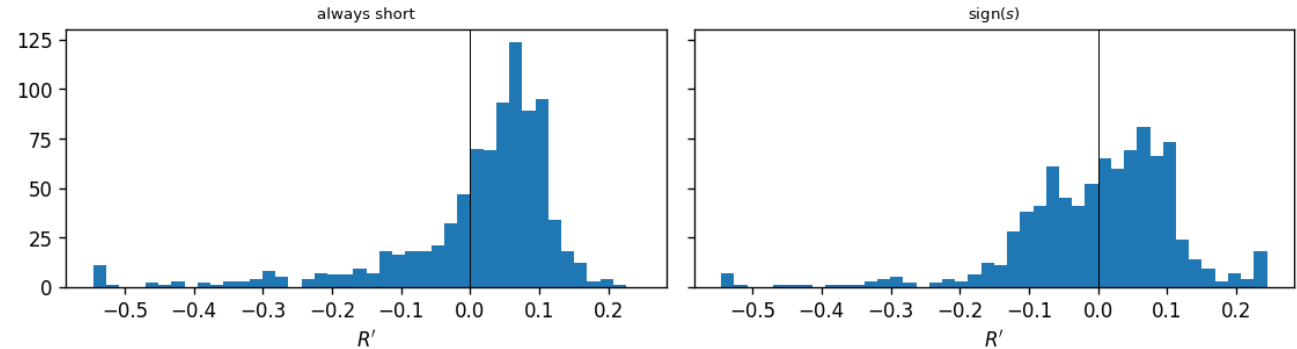
Distribution of the return of the position taken at 10:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 10:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 864 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 10:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.006	0.154	-1.96	0.21	-5.02	43.98	1.18	0.64	0	0.0	-2.44
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.004	0.154	-1.96	1.19	-3.36	43.46	0.81	0.44	289	33.4	-2.66
block-diagonal ridge	865	0.004	0.154	-1.96	1.30	-1.73	43.25	0.73	0.39	301	34.8	-2.71
block-diagonal ridge, without the FOMC columns	865	0.006	0.154	-1.96	1.30	-1.74	43.46	1.22	0.66	311	36.0	-2.45
LightGBM [†]	865	0.005	0.154	-1.96	1.19	-3.01	43.47	0.89	0.48	295	34.1	-2.62
XGBoost [†]	865	0.005	0.154	-1.96	1.19	-3.11	43.56	1.02	0.55	287	33.2	-2.54
lasso (causally tuned) [†]	865	0.003	0.154	-1.96	1.30	-1.75	43.22	0.65	0.35	310	35.8	-2.75
lasso ($\alpha = 10^{-4}$)	865	0.006	0.154	-1.96	1.30	-1.79	43.46	1.21	0.65	287	33.2	-2.46
elastic net (causally tuned) [†]	865	0.005	0.154	-1.96	1.30	-1.81	43.34	0.93	0.50	297	34.3	-2.61

block-diagonal ridge, midpoint fills: entry 10:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



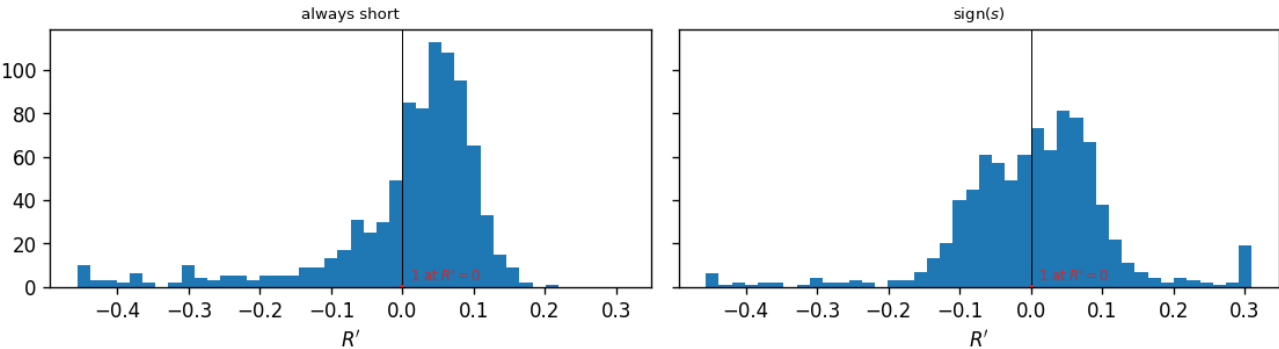
Distribution of the return of the position taken at 10:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 10:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 11:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.004	0.131	-1.27	0.21	-3.41	18.52	0.86	0.46	0	0.0	-3.21
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	865	0.008	0.131	-0.79	1.27	0.55	18.09	1.84	0.99	364	42.1	-2.75
block-diagonal ridge	865	0.005	0.131	-0.81	1.27	0.05	18.12	1.04	0.56	365	42.2	-3.18
block-diagonal ridge, without the FOMC columns	865	0.003	0.131	-0.81	1.27	0.09	18.10	0.66	0.36	378	43.7	-3.39
LightGBM [†]	865	0.007	0.131	-0.81	1.27	-0.04	18.19	1.55	0.84	362	41.8	-2.91
XGBoost [†]	865	0.006	0.131	-0.79	1.27	0.52	18.06	1.24	0.67	370	42.8	-3.06
lasso (causally tuned) [†]	865	0.005	0.131	-0.81	1.27	0.05	18.12	1.03	0.56	379	43.8	-3.17
lasso ($\alpha = 10^{-4}$)	865	0.004	0.131	-0.81	1.27	0.05	18.12	0.94	0.51	362	41.8	-3.24
elastic net (causally tuned) [†]	865	0.005	0.131	-0.81	1.27	0.00	18.13	1.07	0.58	374	43.2	-3.14

block-diagonal ridge, midpoint fills: entry 11:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



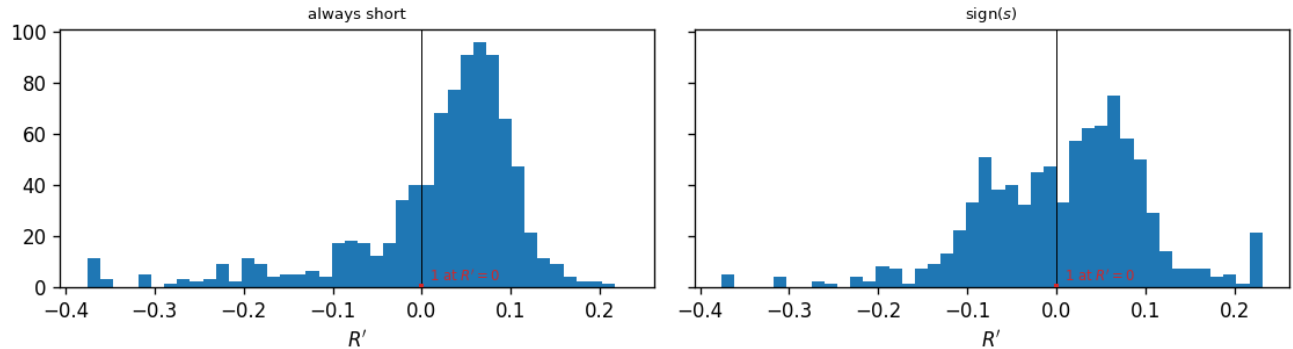
Distribution of the return of the position taken at 11:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 11:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 11:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	865	0.016	0.114	-1.06	0.21	-3.37	19.30	4.05	2.19	0	0.0	-2.13
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	865	0.010	0.115	-0.90	1.06	0.21	16.99	2.68	1.44	333	38.5	-2.93
block-diagonal ridge	865	0.012	0.115	-0.65	1.06	1.32	16.60	2.98	1.61	336	38.8	-2.80
block-diagonal ridge, without the FOMC columns	865	0.013	0.115	-0.65	1.06	1.34	16.62	3.32	1.79	349	40.3	-2.62
LightGBM [†]	865	0.011	0.115	-0.65	1.06	1.27	16.62	2.92	1.57	333	38.5	-2.82
XGBoost [†]	865	0.011	0.115	-0.65	1.06	1.23	16.62	2.69	1.45	334	38.6	-2.93
lasso (causally tuned) [†]	865	0.011	0.115	-0.65	1.06	1.29	16.60	2.73	1.47	344	39.8	-2.93
lasso ($\alpha = 10^{-4}$)	865	0.011	0.115	-0.65	1.06	1.33	16.59	2.90	1.56	351	40.6	-2.84
elastic net (causally tuned) [†]	865	0.013	0.115	-0.65	1.06	1.33	16.62	3.36	1.81	347	40.1	-2.60

block-diagonal ridge, midpoint fills: entry 11:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



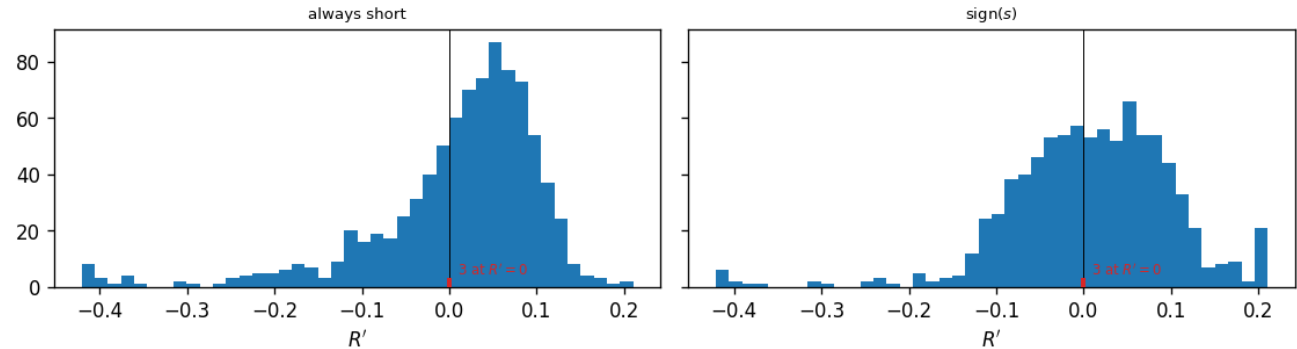
Distribution of the return of the position taken at 11:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 11:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 12:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.008	0.112	-1.02	0.22	-3.16	17.88	2.09	1.12	0	0.0	-3.42
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.007	0.112	-1.02	0.54	-2.01	17.44	1.83	0.98	364	42.0	-3.56
block-diagonal ridge	866	0.008	0.112	-1.02	0.54	-2.10	17.58	2.09	1.13	367	42.4	-3.43
block-diagonal ridge, without the FOMC columns	866	0.007	0.112	-1.02	0.54	-2.09	17.51	1.94	1.04	379	43.8	-3.52
LightGBM [†]	866	0.010	0.112	-1.02	0.54	-1.94	17.80	2.68	1.45	360	41.6	-3.14
XGBoost [†]	866	0.007	0.112	-1.02	0.54	-1.91	17.42	1.83	0.99	372	43.0	-3.59
lasso (causally tuned) [†]	866	0.007	0.112	-1.02	0.54	-2.06	17.46	1.84	0.99	385	44.5	-3.58
lasso ($\alpha = 10^{-4}$)	866	0.009	0.112	-1.02	0.54	-2.12	17.70	2.34	1.26	372	43.0	-3.31
elastic net (causally tuned) [†]	866	0.007	0.112	-1.02	0.54	-2.07	17.43	1.77	0.96	382	44.1	-3.61

block-diagonal ridge, midpoint fills: entry 12:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



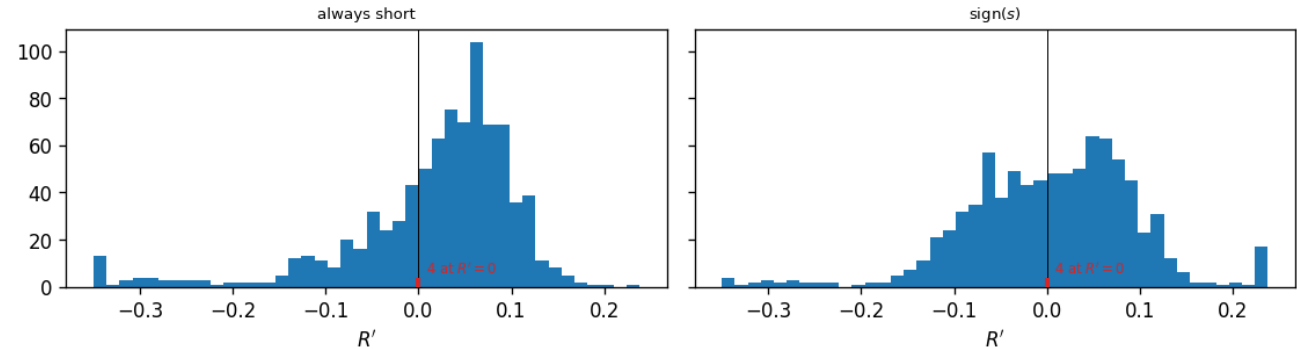
Distribution of the return of the position taken at 12:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 12:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 12:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.013	0.115	-1.23	0.23	-3.74	25.06	3.29	1.78	0	0.0	-2.88
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.007	0.116	-1.23	0.78	-0.64	23.10	1.68	0.91	379	43.8	-3.75
block-diagonal ridge	866	0.005	0.116	-1.23	0.78	-1.21	23.08	1.19	0.64	369	42.6	-4.01
block-diagonal ridge, without the FOMC columns	866	0.005	0.116	-1.23	0.78	-1.18	23.11	1.33	0.72	395	45.6	-3.94
LightGBM [†]	866	0.001	0.116	-1.23	0.78	-1.28	22.87	0.36	0.19	385	44.5	-4.45
XGBoost [†]	866	0.002	0.116	-1.23	0.78	-1.28	22.90	0.45	0.25	391	45.2	-4.40
lasso (causally tuned) [†]	866	0.002	0.116	-1.23	0.68	-1.85	22.93	0.46	0.25	406	46.9	-4.39
lasso ($\alpha = 10^{-4}$)	866	0.004	0.116	-1.23	0.78	-1.20	23.04	1.05	0.57	371	42.8	-4.09
elastic net (causally tuned) [†]	866	0.002	0.116	-1.23	0.68	-1.89	22.92	0.40	0.22	389	44.9	-4.42

block-diagonal ridge, midpoint fills: entry 12:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



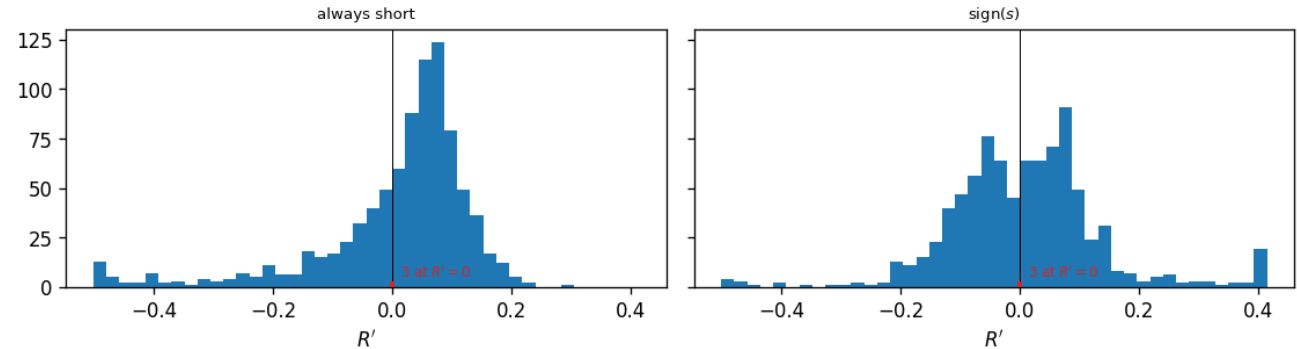
Distribution of the return of the position taken at 12:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 12:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 13:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.002	0.168	-1.72	0.29	-4.23	29.08	0.43	0.23	0	0.0	-3.12
<i>sign(<i>s</i>) (±1 on the sign of <i>s</i>)</i>												
baseline (HAR + calendar OLS)	866	0.012	0.168	-1.72	1.49	0.94	28.85	2.13	1.15	446	51.5	-2.26
block-diagonal ridge	866	0.010	0.168	-1.72	1.49	0.78	28.84	1.77	0.96	418	48.3	-2.45
block-diagonal ridge, without the FOMC columns	866	0.008	0.168	-1.72	1.49	0.50	28.85	1.41	0.76	435	50.2	-2.64
LightGBM [†]	866	0.009	0.168	-1.72	1.49	0.85	28.80	1.54	0.83	447	51.6	-2.57
XGBoost [†]	866	0.008	0.168	-1.72	1.49	0.87	28.78	1.35	0.73	452	52.2	-2.67
lasso (causally tuned) [†]	866	0.009	0.168	-1.72	1.49	0.80	28.81	1.58	0.85	435	50.2	-2.55
lasso ($\alpha = 10^{-4}$)	866	0.011	0.168	-1.72	1.49	0.78	28.85	1.85	1.00	421	48.6	-2.40
elastic net (causally tuned) [†]	866	0.008	0.168	-1.72	1.49	0.79	28.79	1.34	0.72	438	50.6	-2.68

block-diagonal ridge, midpoint fills: entry 13:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



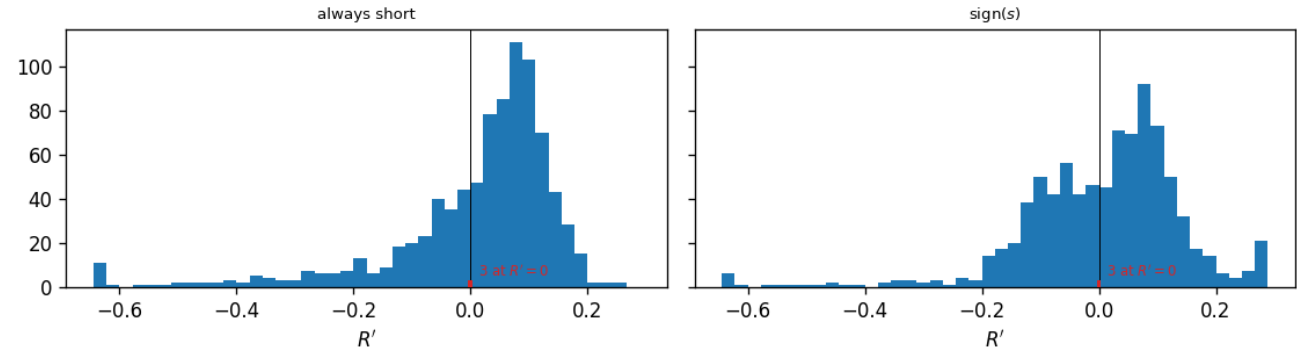
Distribution of the return of the position taken at 13:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 13:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 13:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.008	0.161	-1.07	0.25	-2.85	11.71	1.46	0.79	0	0.0	-3.47
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.005	0.161	-1.07	0.96	-0.86	11.23	0.97	0.52	306	35.3	-3.83
block-diagonal ridge	866	0.010	0.161	-1.01	1.07	-0.61	11.34	1.89	1.02	270	31.2	-3.35
block-diagonal ridge, without the FOMC columns	866	0.010	0.161	-1.01	1.07	-0.39	11.27	1.83	0.99	292	33.7	-3.39
LightGBM [†]	866	0.004	0.161	-1.07	0.85	-1.44	11.24	0.70	0.38	300	34.6	-3.94
XGBoost [†]	866	0.004	0.161	-1.07	0.85	-1.43	11.25	0.74	0.40	303	35.0	-3.91
lasso (causally tuned) [†]	866	0.006	0.161	-1.01	1.07	-0.37	11.17	1.03	0.56	302	34.9	-3.81
lasso ($\alpha = 10^{-4}$)	866	0.010	0.161	-1.01	1.07	-0.61	11.32	1.81	0.98	267	30.8	-3.39
elastic net (causally tuned) [†]	866	0.004	0.161	-1.07	0.96	-1.07	11.19	0.64	0.35	295	34.1	-3.99

block-diagonal ridge, midpoint fills: entry 13:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



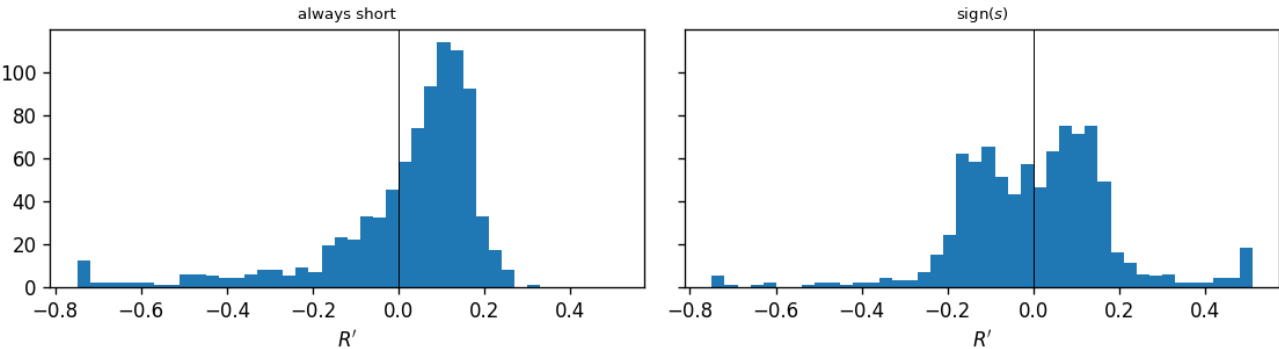
Distribution of the return of the position taken at 13:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 13:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 14:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.008	0.237	-2.65	0.32	-4.48	32.87	0.93	0.50	0	0.0	-2.62
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.006	0.237	-2.65	1.63	-1.70	32.45	0.75	0.40	451	52.1	-2.77
block-diagonal ridge	866	0.007	0.237	-2.65	1.63	-1.61	32.50	0.91	0.49	427	49.3	-2.68
block-diagonal ridge, without the FOMC columns	866	0.005	0.237	-2.65	1.63	-1.58	32.42	0.68	0.36	442	51.0	-2.81
LightGBM [†]	866	0.001	0.237	-2.65	1.63	-1.49	32.26	0.11	0.06	469	54.2	-3.11
XGBoost [†]	866	-0.000	0.237	-2.65	1.63	-1.52	32.24	-0.01	-0.01	472	54.5	-3.18
lasso (causally tuned) [†]	866	0.000	0.237	-2.65	1.63	-1.69	32.24	0.01	0.01	450	52.0	-3.16
lasso ($\alpha = 10^{-4}$)	866	0.009	0.237	-2.65	1.63	-1.62	32.57	1.10	0.59	424	49.0	-2.58
elastic net (causally tuned) [†]	866	-0.001	0.237	-2.65	1.63	-1.55	32.22	-0.07	-0.04	446	51.5	-3.22

block-diagonal ridge, midpoint fills: entry 14:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



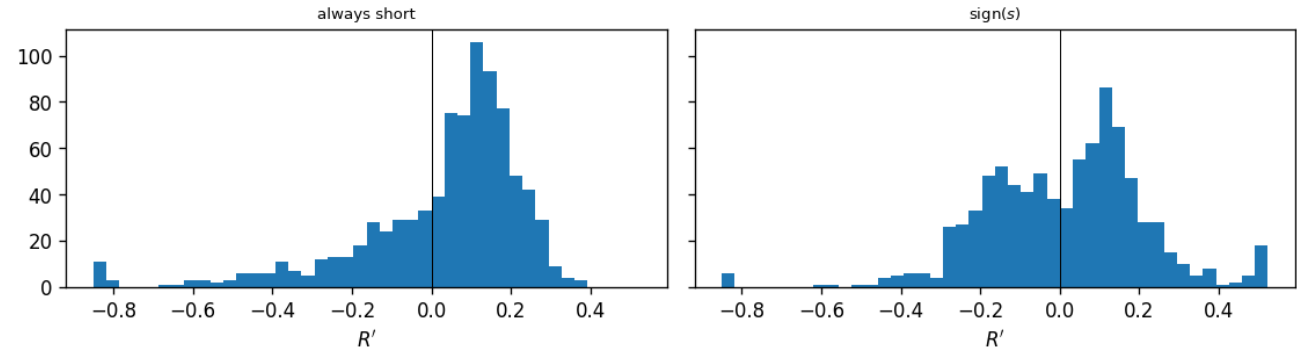
Distribution of the return of the position taken at 14:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 14:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 14:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.019	0.300	-5.15	0.38	-7.56	108.09	1.86	1.00	0	0.0	-1.52
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.007	0.300	-5.15	2.01	-5.13	105.90	0.67	0.36	335	38.7	-2.19
block-diagonal ridge	866	0.011	0.300	-5.15	2.01	-5.45	106.38	1.06	0.57	342	39.5	-1.98
block-diagonal ridge, without the FOMC columns	866	0.011	0.300	-5.15	2.01	-5.12	106.33	1.06	0.57	350	40.4	-1.98
LightGBM [†]	866	0.005	0.300	-5.15	2.01	-5.44	105.73	0.48	0.26	350	40.4	-2.29
XGBoost [†]	866	0.006	0.300	-5.15	2.01	-5.40	105.86	0.60	0.32	367	42.4	-2.23
lasso (causally tuned) [†]	866	0.007	0.300	-5.15	2.01	-5.41	105.97	0.71	0.38	361	41.7	-2.17
lasso ($\alpha = 10^{-4}$)	866	0.009	0.300	-5.15	2.01	-5.43	106.21	0.92	0.50	337	38.9	-2.06
elastic net (causally tuned) [†]	866	0.009	0.300	-5.15	2.01	-5.43	106.18	0.89	0.48	358	41.3	-2.07

block-diagonal ridge, midpoint fills: entry 14:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



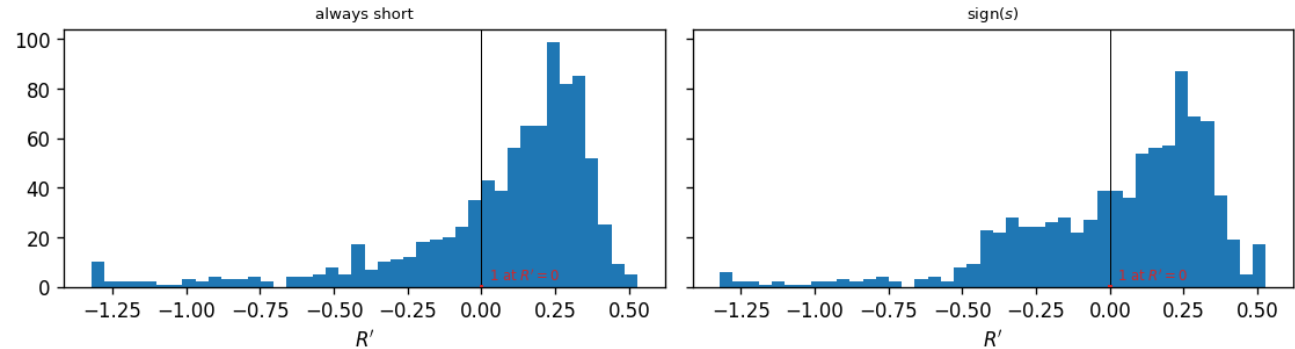
Distribution of the return of the position taken at 14:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 14:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 15:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe _{clk} [×]
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.060	0.379	-3.31	0.58	-3.03	15.14	4.64	2.51	0	0.0	0.18
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)												
baseline (HAR + calendar OLS)	866	0.026	0.383	-3.31	2.85	-1.16	13.02	2.01	1.08	201	23.2	-1.20
block-diagonal ridge	866	0.033	0.383	-3.31	2.85	-0.97	13.11	2.52	1.36	186	21.5	-0.93
block-diagonal ridge, without the FOMC columns	866	0.032	0.383	-3.31	2.85	-0.96	13.08	2.45	1.32	189	21.8	-0.97
LightGBM [†]	866	0.024	0.383	-3.31	2.85	-1.08	12.96	1.86	1.00	205	23.7	-1.28
XGBoost [†]	866	0.022	0.383	-3.31	2.85	-1.07	12.92	1.69	0.91	220	25.4	-1.37
lasso (causally tuned) [†]	866	0.033	0.382	-3.31	2.85	-0.84	13.07	2.55	1.37	198	22.9	-0.92
lasso ($\alpha = 10^{-4}$)	866	0.034	0.382	-3.31	2.85	-0.97	13.13	2.61	1.41	185	21.4	-0.89
elastic net (causally tuned) [†]	866	0.029	0.383	-3.31	2.85	-1.19	13.09	2.22	1.20	190	21.9	-1.09

block-diagonal ridge, midpoint fills: entry 15:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



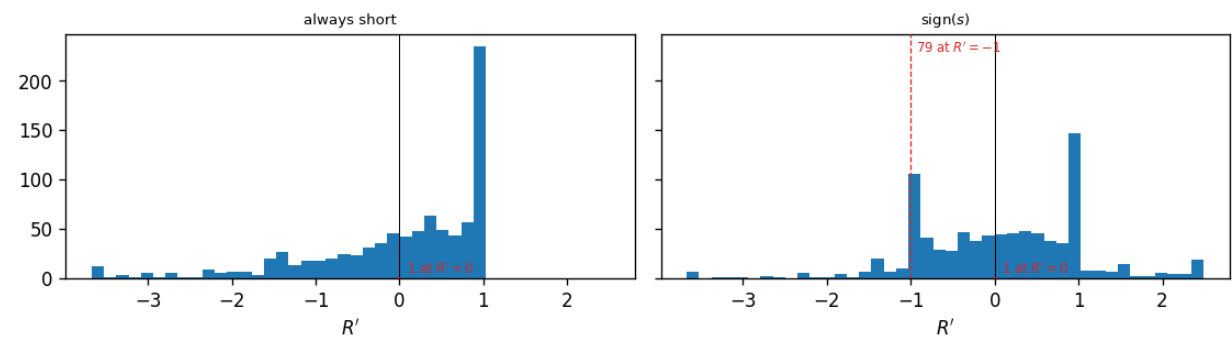
Distribution of the return of the position taken at 15:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 15:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 15:30 ET, cash-settled at the close

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{clk}	<i>n</i> _{buy}	%buy	Sharpe [×] _{clk}
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.014	1.128	-10.32	1.00	-2.31	10.89	0.38	0.20	0	0.0	-0.26
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.069	1.126	-5.42	10.32	0.58	10.71	1.79	0.97	286	33.0	0.50
block-diagonal ridge	866	0.095	1.124	-5.42	10.32	0.92	10.61	2.48	1.34	346	40.0	0.87
block-diagonal ridge, without the FOMC columns	866	0.090	1.124	-5.42	10.32	1.00	10.59	2.36	1.27	355	41.0	0.80
LightGBM [†]	866	0.103	1.123	-5.42	10.32	1.01	10.58	2.71	1.46	290	33.5	0.99
XGBoost [†]	866	0.106	1.123	-5.42	10.32	0.91	10.62	2.78	1.50	279	32.2	1.03
lasso (causally tuned) [†]	866	0.100	1.123	-6.84	10.32	0.46	10.78	2.62	1.41	321	37.1	0.94
lasso ($\alpha = 10^{-4}$)	866	0.107	1.123	-5.42	10.32	1.22	10.50	2.80	1.51	358	41.3	1.04
elastic net (causally tuned) [†]	866	0.072	1.125	-6.84	10.32	0.46	10.74	1.90	1.02	307	35.5	0.56

block-diagonal ridge, midpoint fills: entry 15:30 ET, cash-settled at the close; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



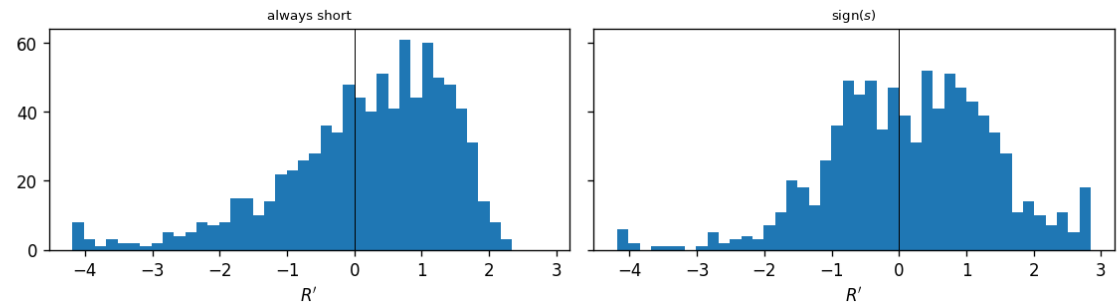
Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the package expires worthless).

One trade a day: enter the nearest-OTM package at 15:30 ET and let it cash-settle at the official close. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

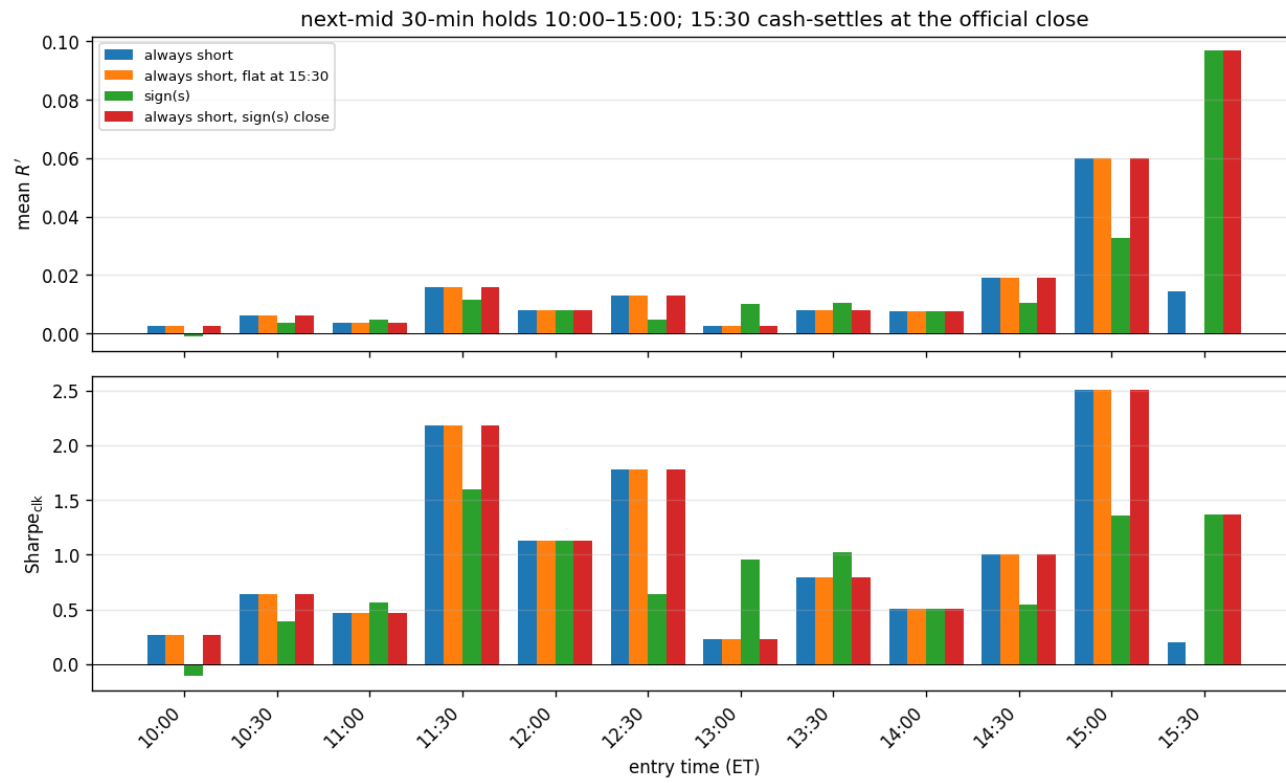
Rule table by strategy — pooled over the twelve entry windows

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	<i>n</i> _{buy}	%buy	Sharpe ^x _{ann}
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.160	1.338	-11.40	2.24	-1.92	8.73	3.53	1.90	0	0.0	-1.92
<i>sign(s) (±1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.163	1.356	-5.96	11.70	0.27	7.40	3.54	1.91	266	30.7	-2.35
block-diagonal ridge	866	0.198	1.355	-6.26	11.40	0.55	7.39	4.30	2.32	252	29.1	-1.94
block-diagonal ridge, without the FOMC columns	866	0.192	1.348	-6.15	11.40	0.60	7.41	4.19	2.26	267	30.8	-2.02
LightGBM [†]	866	0.179	1.347	-5.96	11.40	0.58	7.25	3.92	2.12	277	32.0	-2.16
XGBoost [†]	866	0.173	1.357	-5.96	11.40	0.55	7.30	3.75	2.02	271	31.3	-2.21
lasso (causally tuned) [†]	866	0.180	1.337	-6.58	11.70	0.37	8.07	3.96	2.13	276	31.9	-2.17
lasso ($\alpha = 10^{-4}$)	866	0.209	1.344	-6.26	10.65	0.63	6.23	4.59	2.47	248	28.6	-1.82
elastic net (causally tuned) [†]	866	0.151	1.361	-6.72	11.40	0.31	7.19	3.26	1.76	266	30.7	-2.47

block-diagonal ridge, midpoint fills: daily sum over the twelve entry windows 10:00-15:30 ET; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)

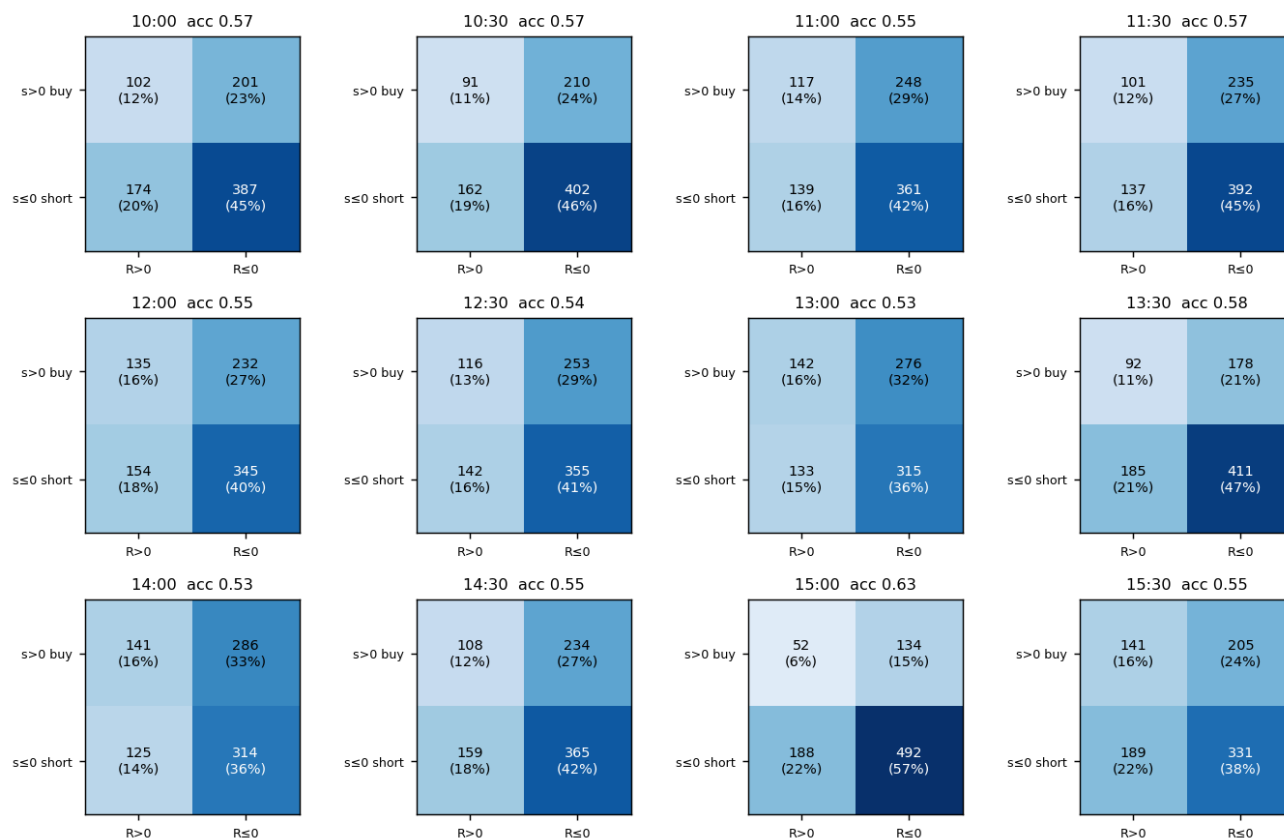


Distribution of the return of the daily sum over the twelve entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and Sharpe_{ckpt} by entry clock (grouped bars). Sharpe_{ckpt} is that clock's daily series $\times \sqrt{252}$, not the pooled Sharpe_{ann}. Next-mid 30-minute holds 10:00–15:00; 15:30 cash-settles at the official close.

sign(s) vs sign of long-package R (mid), block-diagonal ridge, 866 days



Confusion matrix per entry clock: sign(s) (buy if $s > 0$) against the sign of the long-package midpoint return. 15:30 is the paper trade.

Each row is the daily sum over the twelve entry windows 10:00–15:30 ET: one number per expiration day, then the usual statistics on that daily series. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Notes

One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one 30-minute hold per expiration day. It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s twelve holds. That reserved name is the book’s daily P&L after pooling the session. At 15:30 the two coincide (one trade that day); they do not on 10:00–15:00 or on the pooled page. $\text{Sharpe}^\times$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^\times$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $RV_t / \sum_{s \geq t} RV_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.